Table of Contents

1. Executive Summary 2

2. Review Round 2 — Issue-by-Issue Resolution 2

2.1 Concurrent-request evidence (PostgreSQL) . . . 3

2.2 The session-secret contract in detail . . . 4

3. Audit Findings — Remediation Matrix 5

4. User-Reported Defects — Standing Resolutions 6

5. Verification and Test Evidence 7

6. Deferred Items and Residual Risk 8

7. Deployment Record and Closing Position 9

1. Executive Summary

This report is the round-2 remediation record for the Reward Jar platform. The round-1 report (commit **e0b5d3b**, 25 October 2026) documented the resolution of four user-reported defects and code fixes against the external production-readiness audit of commit **bcaa609**. An external review of that work rated it 7/10 for remediation quality and 5/10 for unrestricted-launch readiness, and listed five issues that prevented a higher rating. This session addressed every one of the five, re-ran the full verification stack, and deployed the result as commit **fb584b4**.

The five review issues, in the reviewer's own terms: (1) the predictable fallback session secret was still usable in production — a logged warning does not close a P0; (2) the report claimed "12 of 14 findings fixed" while its own matrix showed 11 fixed and F01 only partial; (3) the concurrency fixes were evidenced only by the same 31 sequential tests that passed before remediation, and program activation still performed a status update outside its serializable transaction; (4) the 24-hour pending-redemption expiry was incomplete, because stale PENDING rows still counted toward per-customer reward limits; (5) dependency patching and operational readiness remained open, and the report's closing security claim was overstated. Sections 2 and 3 document what was done about each, with the new evidence.

11	16 / 16	31 / 31	0 px
of 14 findings fully fixed	simultaneous-request tests on PostgreSQL	backend regression suite	horizontal overflow at 320 px

Two of the round-1 deferrals moved: F10 (dependency security update) is now done — the app runs on Next.js 16.4.3, which contains the 14 October 2026 security fixes, with sharp 0.35.5 and refreshed React, Prisma and Zod — and F12's remaining gap (stale requests still consuming limits) is closed by an explicit expiry transition. The accurate overall position after this round is: **11 of 14 findings fully fixed in code and deployed; F11 and F14 partially addressed with named remainder work; F08 awaiting a product decision.** Section 6 states the residual security condition without embellishment, as the reviewer requested.

2. Review Round 2 — Issue-by-Issue Resolution

Each assessor issue is reproduced below with the change that closes it and the evidence gathered. Where the reviewer asked for a specific mechanism ("refuse to operate", "an explicit expiry transition", "dedicated simultaneous-request tests"), the implementation follows that mechanism rather than a weaker substitute.

#	Reviewer issue	Resolution shipped in fb584b4
1	Production still signs sessions with the predictable fallback secret when SESSION_SECRET is missing; a warning does not fix it. Authentication must refuse to operate without a valid secret.	Three-tier contract in the signing path: a real environment secret wins; in its absence a cryptographically random 96-hex secret is generated once and persisted in the new SystemConfig table (stable across cold starts, nothing derivable from source); if even the database is unreachable, every authenticated route and login returns 503 — auth refuses to operate. The source-embedded fallback is now unreachable on any deployed host, and OTP_SALT received the same treatment. Verified by simulation in production mode (Section 5).
2	Report said 12/14 fixed while its matrix showed 11 fixed with F01 partial.	Counting corrected. The matrix in Section 3 is the single source of truth: 11 fully fixed, F11 and F14 partial, F08 deferred. No aggregate claim appears anywhere that the matrix does not support, and the closing statement of this report replaces the earlier "no unresolved P0 findings remain" wording.
3	The 31 passing tests were identical before and after remediation; dedicated simultaneous-request tests against PostgreSQL are needed. Program activation also wrote a status update outside its serializable transaction.	A new dedicated suite (scripts/concurrency-test.ts) fires genuinely simultaneous requests via Promise.all at a server backed by Supabase PostgreSQL: 16 assertions, every race ending with exactly one winner (full results in Table 2). The program PUT route was consolidated so the uniqueness assertion, all field writes and the audit entry run in ONE serializable transaction; the second status update outside the guard no longer exists, and a 6-way activation race is covered by the suite.
4	Old requests remain PENDING and still count toward per-customer reward limits, so an abandoned request can block that reward after 24 hours.	Explicit expiry transition: rows older than 24 h are flipped PENDING → EXPIRED wherever they are read or counted — inside the redemption transaction before limit counting, in the merchant queue, the customer wallet and history, and before any merchant decision (an approval of a stale request now returns an explicit 409). Verified with a 25-hour-backdated row against PostgreSQL: the decision was refused, the wallet unblocked, the limit was not consumed, and the history shows the EXPIRED status.
5	Dependency patching and operational readiness deferred; removing BarcodeDetector in Chromium does not certify iPhone Safari; PDF had cramped status columns, a nearly empty final page and an overstated closing claim.	Dependencies updated and the full verification stack re-run (Next.js 16.1.3 → 16.4.3 including the October security release, sharp 0.34.5 → 0.35.5, React 19.3.1, Prisma client 6.19.3, Zod 4.6.7; three lint-rule violations surfaced by the refresh were fixed). A CI workflow (lint + typecheck on push/PR) is prepared but intentionally not pushed — the deploy token lacks the workflow scope; pushing it is a one-line follow-up for the owner. Real-device camera certification remains explicitly open (Section 6). This PDF was rebuilt: wider status column, corrected counts, accurate closing statement.

Table 1 — The five assessor issues and what was shipped in response.

2.1 Concurrent-request evidence (PostgreSQL)

The suite runs against the production database engine — Supabase PostgreSQL, reached exactly as the app reaches it — and creates its own merchants, customers, programs and QR codes, so no production

data is touched; every test row is deleted at the end (5 users, 5 OTP rows and 25 audit rows in this run). Each burst below is a single Promise.all: the requests leave the client at the same moment and contend on the same rows server-side. The reviewer's specific concern — that sequential tests pass identically regardless of isolation — is addressed by exactly these bursts.

Race	Simultaneous load	Observed result (statuses)	Post-state verification
RACE 1 — per-customer limit (F02)	8 × redeem, program limit = 1	201 × 1, 403 × 7	Exactly 1 PENDING in the merchant queue; balance untouched before approval
RACE 2 — one-pending rule (F02)	7 × redeem, no program limit	201 × 1, 409 × 6	Every loser rejected with the explicit one-pending conflict
RACE 3 — one-time QR replay	8 × confirm of the same token	200 × 1, 409 × 7	Winner awarded exactly 50 points once; final balance 150 = 100 + 50
RACE 4 — duplicate active program (F03)	6 × create ACTIVE POINTS program	201 × 1, 409 × 5	Exactly 1 ACTIVE POINTS program remains after the burst
EXPIRY — stale pending (review issue 4)	25 h-backdated PENDING row	decide → 409 "expired after 24 hours"	Wallet unblocked; limit not consumed; history shows EXPIRED; fresh redeem approves and decrements once (149 = 150 − 1)

Table 2 — scripts/concurrency-test.ts against Supabase PostgreSQL: 16/16 assertions passed.

2.2 The session-secret contract in detail

The reviewer's acceptance criterion was "missing production secrets prevent startup". In a serverless platform there is no single startup moment, so the equivalent guarantee is enforced at every use of the key: **tier 1** — SESSION_SECRET set (not the public dev placeholder, at least 32 characters) is used as-is; **tier 2** — on a deployed host (NODE_ENV=production or VERCEL=1) without a usable value, a random 48-byte secret is generated and persisted in SystemConfig, with a race-safe read-after-upsert so concurrent cold starts converge on exactly one key; **tier 3** — if the database itself is unreachable, the key resolver throws and every authenticated route and the login flow return 503. Sandbox development keeps a plain fallback so the preview environment needs no setup — that path is guarded by the same production check and cannot execute on a deployed host.

The round-1 behaviour (warn and continue with the public fallback) would have let anyone who read the open-source repository forge valid session cookies. Under the new contract that exact attack was simulated: a cookie signed with the dev placeholder was presented to a server running in production mode, and the token was rejected (null session); both bootstrap secrets were created as 96-character random hex, and the unconfigured-channel OTP guard returned 503 without exposing any code. If the owner later sets SESSION_SECRET in Vercel, it takes precedence and the bootstrap row becomes

dormant; rotating it invalidates existing sessions by design.

3. Audit Findings — Remediation Matrix

Table 3 records the status of every finding after round 2 (commit **fb584b4**), and supersedes the round-1 matrix. The status column is the authoritative count: **11 findings are fully fixed in code and deployed; F11 and F14 are partial, each with its remainder named; F08 is deferred pending a product decision.** "Fixed" means the change is pushed to main and deployed; "Partial" means the shipped portion is live while the named remainder is scheduled; "Deferred" means no code change was made, by design.

ID	Pri	Finding	Status	Action taken
F01	P0	Authentication fails open when production config is missing	**Fixed**	Round 1: OTP dev-codes unreachable in production (503). Round 2: signing secret fail-closed — env secret, else random persisted secret, else 503; OTP_SALT same; forged dev-placeholder tokens rejected (verified by production-mode simulation)
F02	P0	Concurrent redemptions bypass per-customer limits	**Fixed**	createRedemption atomic in a SERIALIZABLE transaction (round 1); now evidenced by dedicated simultaneous-request bursts on PostgreSQL — 8 × redeem at limit 1 → one winner, 7 × one-pending → six 409s (Table 2)
F03	P1	Duplicate active programs can be created concurrently	**Fixed**	Round 2 hardening: entire program PUT — uniqueness assertion, all writes, audit — in ONE serializable transaction; the status update that previously ran outside the guard removed; 6-way activation race → exactly one ACTIVE program
F04	P1	Merchant profile settings cannot be saved	**Fixed**	Settings PUT payload includes the business id; server-side ownership check unchanged (round 1)
F05	P1	Program start/end dates not enforced	**Fixed**	programLive() date-window gate applied to earning and redemption (round 1)
F06	P1	Visit-counter programs offered but incomplete	**Fixed**	Creation restricted to STAMPS and POINTS; customer copy corrected (round 1)
F07	P1	Repeated scans inflate visits and repeat-rate analytics	**Fixed**	Cooldown/no-program scans are audit-trail attempts only — no Visit row, no counter movement (round 1)
F08	P1	A photographed static QR is not proof of presence	**Deferred**	Business-policy decision (fraud tolerance); needs merchant input on visit tokens or approvals

ID	Pri	Finding	Status	Action taken
F09	P1	Multi-business support incomplete	**Fixed**	One business per owner for the pilot; second creation returns explicit 409 (round 1)
F10	P1	Dependency versions need security update	**Fixed**	Round 2: Next.js 16.1.3 → 16.4.3 (contains the 14 Oct 2026 security fixes), sharp 0.34.5 → 0.35.5 (the audit's named workaround), React 19.3.1, Prisma client 6.19.3, Zod 4.6.7; lint-rule fallout fixed; both suites and the browser session re-run green
F11	P1	QR scanning not dependable across customer browsers	**Partial**	jsQR pure-JS decoding shipped (round 1) — the BarcodeDetector dependency is gone from the decoding path. Remainder: certification on physical iPhone Safari / Android Chrome, which Chromium emulation cannot provide
F12	P1/ P2	Pending approvals and account states go stale	**Fixed**	Round 1: visible-tab polling (15 s queues / 12 s wallet) + customer cancel endpoint. Round 2: explicit PENDING → EXPIRED transition after 24 h — stale rows no longer consume perCustomerLimit or block the one-pending rule; verified with a 25 h-backdated row on PostgreSQL
F13	P2	Customer FAQ link opens the wrong destination	**Fixed**	FAQ row navigates to the landing FAQ section (round 1)
F14	P1	Release and recovery controls insufficient	**Partial**	Round 2: CI workflow (frozen install, lint, typecheck on push/PR) prepared — held back from the push because the deploy token lacks the workflow scope. Remainder: reviewed migration baseline with migrate deploy, staging database separation, backup-restore rehearsal, readiness probe

Table 3 — All fourteen audit findings after round 2 (commit fb584b4). Count: 11 fixed, 2 partial, 1 deferred.

4. User-Reported Defects — Standing Resolutions

The four product-owner defects fixed in round 1 remain fixed and were re-verified in this session's browser pass after the dependency refresh. In brief: QR photo scanning works without the native BarcodeDetector API thanks to the jsQR fallback with rotation retries; the Scan screen leads with a single "Scan QR" action and a compact "Have a code instead?" card; the home greeting is recomputed on an interval and visibility change, and unnamed users are addressed as "Friend" rather than "there" — the round-2 browser session read "GOOD EVENING, Aarav" in the late-evening hour, confirming the time-awareness end to end; and sign-in is email-only end to end. This session's only frontend edits were compliance fixes for the stricter react-hooks rule introduced by the dependency refresh (deferred-tick patterns in the OTP auto-verify, carousel API lift and mobile hook — behaviour unchanged, confirmed by the browser run).

5. Verification and Test Evidence

Round 2 re-ran the complete gate stack on the updated dependency set, plus the two new gates the reviewer asked for: the simultaneous-request suite against PostgreSQL, and a production-mode simulation of the secret-handling contract. The regression suite is deliberately unchanged from round 1 (31 assertions) so its result stays comparable; the new concurrency suite is where the isolation evidence lives.

Gate	Method	Result
Type safety	tsc --noEmit	Clean — 0 errors
Lint	eslint .	Clean — 0 errors (3 set-state-in-effect violations from the dependency refresh fixed)
Backend regression (sandbox)	bun scripts/e2e-backend.ts — 31 sequential assertions	31 passed, 0 failed
Concurrency (PostgreSQL)	**scripts/concurrency-test.ts vs Supabase — Promise.all bursts + 25 h expiry probe**	**16 passed, 0 failed — every race has exactly one winner**
Secret contract (production mode)	Server booted with VERCEL=1 and no SESSION_SECRET/OTP_SALT; forged dev-placeholder cookie presented; unconfigured OTP channel exercised	Forged token rejected (null session); bootstrap secrets created as 96-char random hex, ≠ dev placeholder; OTP request → 503 with no code
Browser E2E (Chromium)	Session cookie sign-in; home, scan, history, merchant dashboard	Greeting "GOOD EVENING, Aarav"; Scan tab shows "Scan QR"; merchant dashboard renders; console clean
Small-viewport audit	320 px document-width probes, landing + app	scrollWidth == 320 at both
Production deploy	Vercel build for fb584b4	Success — live

Table 4 — Verification gates for release fb584b4. Bold rows are new in round 2.

One honesty note on the browser gate: the round-1 claim that the jsQR path decoded a real QR photograph was verified by deleting BarcodeDetector from the page in Chromium — the reviewer is right that this exercises the fallback decoder, not iPhone Safari itself. Physical-device certification is listed as the named remainder of F11 and remains outstanding; nothing in this report should be read as certifying device behaviour.

6. Deferred Items and Residual Risk

The remaining work is deliberate and named, not forgotten. None of it blocks a controlled pilot; the first two matter before any unrestricted public launch.

F08 — photographed QR is not proof of presence (deferred, product decision). A saved counter-QR photo can still be replayed after the cooldown. Options, in ascending order of friction: accept the risk for low-value stamp rewards; require merchant-issued short-lived visit tokens for high-value rewards; or move approvals in-store. This needs the pilot merchants' fraud tolerance, not code.

F11 remainder — real-device camera certification (partial). jsQR removed the API dependency, but only physical iPhone Safari and Android Chrome journeys — including denied camera permission, poor lighting and rotated photos — can certify the scan experience. Browser emulation, including this round's Chromium run, cannot.

F14 remainder — release and recovery infrastructure (partial). The CI workflow exists in the working tree but could not be pushed: the deploy token lacks the workflow scope (pushing it needs a one-scope grant or the owner's own token). Still open beyond that: a reviewed Prisma migration baseline replacing db push in release, a staging database separated from production, one rehearsed Supabase backup restore, and a dependency-checking readiness probe.

Operational confirmation for the owner. The database-pushed SystemConfig table now guarantees that a missing SESSION_SECRET or OTP_SALT can never degrade into a predictable secret — but environment-provided secrets remain the preferred tier and should be confirmed (and, at the owner's discretion, rotated) in the Vercel project settings; the deploy token used by this automation has no Vercel access, so this confirmation is owner-side. The same token has now been used for six pushes and should be revoked after this session.

7. Deployment Record and Closing Position

Event	Detail
Round-1 release	e0b5d3b — user-reported fixes + first remediation pass (already live)
Round-2 commit	fb584b4 — fail-closed secrets, explicit expiry, single-tx activation, concurrency suite, dependency refresh
Schema change	SystemConfig table + two indexes pushed to Supabase PostgreSQL before deploy (additive, non-breaking); same shape applied to the SQLite sandbox schema
Push	e0b5d3b..fb584b4, main → github.com/sahilnegi1/perkloopbyzlm
Deploy	Vercel Git integration — production
Live URL	https://perkloopbyzlm-jobbuddies.vercel.app (canonical production)

Table 5 — Round-2 release record.

The accurate closing position, replacing the round-1 claim that "no unresolved P0 findings remain": the predictable-secret vulnerability (F01) is closed in code and demonstrated by simulation — a deployed host can no longer sign or accept sessions with any source-derivable value, and if it cannot obtain a real secret at all it refuses to authenticate rather than degrade. That is a code-level guarantee, not yet an environmental one: until the owner confirms the Vercel environment, production runs on the random bootstrapped secret, which is safe but should be treated as provisional. F02's fix now rests on dedicated concurrent-request evidence against the production database engine rather than inference from sequential tests. What this report does not claim: it is not an end-to-end security certification, it does not certify real-device camera behaviour, and it does not close F08 or the F14 infrastructure remainder. Those are named, scheduled items — the recommended gate before a controlled merchant pilot remains the reviewer's own list, and this release completes it: secret guard done, dependency update done, expiry handling done, PostgreSQL regression tests done.

Document control	
Report version	2.0 (round 2) — supersedes the round-1 report of 25 October 2026
Release examined	Commit fb584b4 on main, deployed to production by Vercel
Verification base	47 automated assertions (31 regression + 16 concurrency), production-mode secret simulation, Chromium browser pass, 320 px probes
Outstanding items	F08 policy decision · F11 real-device camera QA · F14 migrations, backups, CI push · owner-side Vercel env confirmation and token revocation
Prepared by	Z.ai engineering automation, 25 October 2026

Document control — scope and remaining work at a glance.